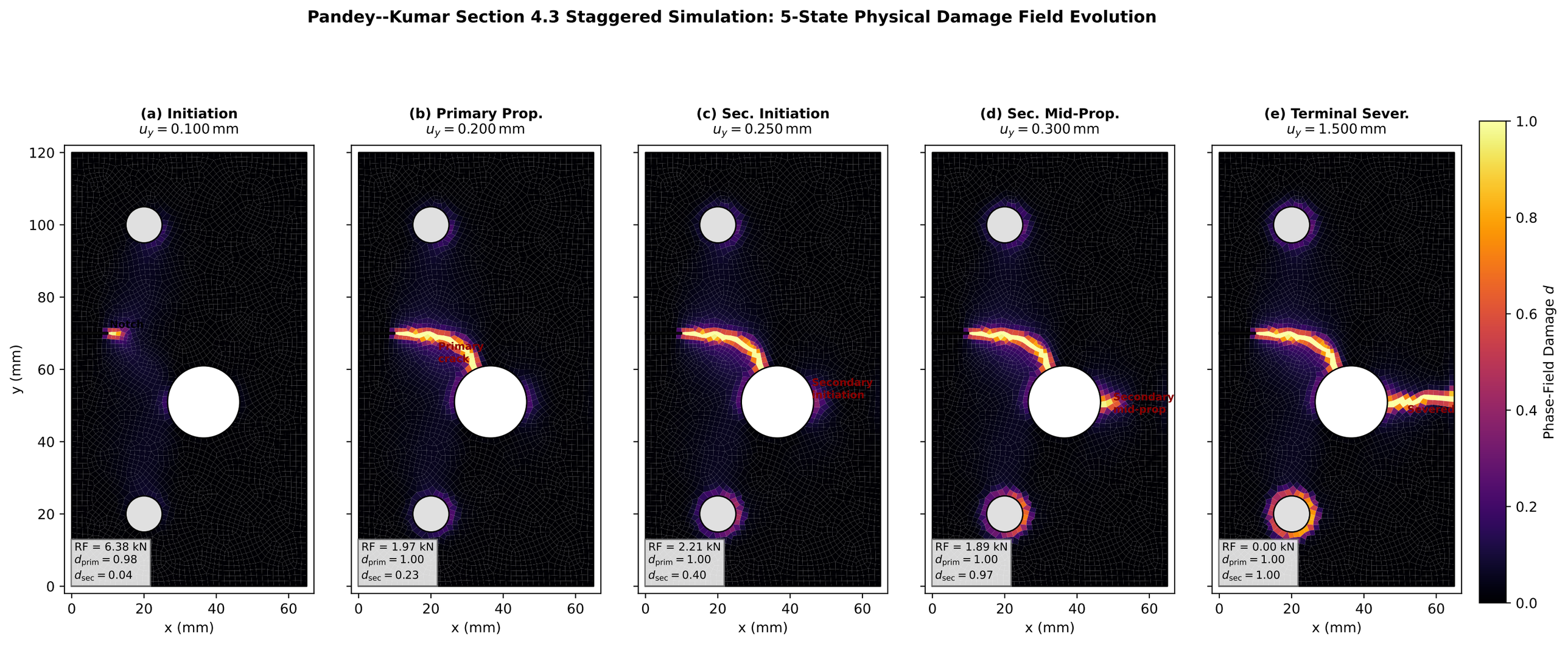

Pandey--Kumar Section 4.3 Staggered Simulation: 5-State Physical Damage Field Evolution
(a) Initiation
$u_y = 0.100$ mm
(b) Primary Prop.
$u_y = 0.200$ mm
(c) Sec. Initiation
$u_y = 0.250$ mm
(d) Sec. Mid-Prop.
$u_y = 0.300$ mm
(e) Terminal Sever.
$u_y = 1.500$ mm
Notch
Primary crack
Secondary initiation
Secondary mid-prop
Severed
RF = 6.38 kN
$d_{prim} = 0.98$
$d_{sec} = 0.04$
RF = 1.97 kN
$d_{prim} = 1.00$
$d_{sec} = 0.23$
RF = 2.21 kN
$d_{prim} = 1.00$
$d_{sec} = 0.40$
RF = 1.89 kN
$d_{prim} = 1.00$
$d_{sec} = 0.97$
RF = 0.00 kN
$d_{prim} = 1.00$
$d_{sec} = 1.00$
y (mm)
x (mm)
Phase-Field Damage $d$
0
20
40
60
80
100
120
0.0
0.2
0.4
0.6
0.8
1.0